LoCoBench: Logical Coherence Score Evaluation

Gold-relation and mined arms on generated items

FactReasoner — LoCoBench evaluation

August 7, 2026

1 Setup

This report evaluates the Logical Coherence Score (LCS) pipeline on the **10 generated items** of `locobench-claude-5-test`. Unlike the mining experiments, relations are *not* mined by a model here: each item already carries its inter-atom relations as gold labels, and those labels are compiled directly into the coherence MRF. No LLM is involved, so every number below is deterministic and reproducible offline (Merlin is the only subprocess).

Atom priors. An atom's unary factor is $[1-\pi_i, \pi_i]$ with $\pi_i = 0.9$ when the item marks the atom `factual` and $\pi_i = 0.1$ when it does not. The coherence MRF therefore starts from the corpus's own factuality labels — the label-driven analogue of the two-stage model, where stage 1's posterior marginals play this role. Hard 1/0 priors are deliberately avoided: they would zero out worlds and make several readouts degenerate.

Edge probabilities. A gold relation carries an intended strength band and a strength range; the factor probability is the range's **midpoint** (strong $[0.85, 1] \to 0.925$, moderate $[0.6, 0.84] \to 0.720$, weak $[0.35, 0.59] \to 0.470$). Because gold is a *label* rather than an estimate, the type confidence $P(\tau \mid a_i, a_j)$ is fixed at 1.0 and the conditional strength is that same midpoint. Any apparent precision beyond the band is an artefact of the midpoint convention, not a measurement.

Resolved concessions. A concession the text resolves is softened by $p \mapsto p\,(1-\lambda)$ with $\lambda = 0.45$ (deep-dive Eq. 2). The resolver is read from the item's own `resolver_atom_id`, so the miner's text heuristic for spotting a holding atom is bypassed entirely — when the label states the resolver, guessing it would be the wrong instrument.

Ordering-only relations. `Precedence` and `Succession` compile to Level-1 `none`: they record source/target order but couple no truth values, so they contribute **no factor**. They are counted in the dataset table and excluded from the MRF, exactly as `lcs.taxonomy.compile_sense` dictates.

Arms. Each item is scored under `gold` (all gold edges) and `gold_valid` (valid edges only) and `mined:llama-3.3-70b-instruct:windowed` (mined by llama-3.3-70b-instruct, windowed) and `mined:llama-3.3-70b-instruct:all_pairs` (mined by llama-3.3-70b-instruct, all pairs). The second arm drops the deliberately-planted invalid relations, which isolates what those planted errors cost. All four readouts are computed for both: `mean_marginal`, `consistency`, `reified`, `log_partition`.

2 Dataset

Table 1: Per-item size. **Gold** is every relation the item lists; **ord-only** are the Precedence/Succession edges that carry no truth coupling and so produce no factor; **MRF** is what remains and is scored; **valid** is the subset whose `validity` is `valid`.

Family	Rung	Name	Atoms	Gold	ord-only	MRF	valid
`f001`	0	worse	16	11	1	10	5
`f001`	1	base	16	10	1	9	5
`f001`	2	concession_resolved	16	10	1	9	5
`f001`	3	fix_one_conflict	16	9	1	8	5
`f001`	4	coherent	16	8	1	7	4
`f002`	0	worse	16	11	1	10	6
`f002`	1	base	16	10	1	9	6
`f002`	2	concession_resolved	16	10	1	9	6
`f002`	3	fix_one_conflict	16	9	1	8	6
`f002`	4	coherent	16	8	1	7	6

Across the dataset there are 160 atoms, of which 140 are marked `factual` (prior 0.9) and 20 are not (prior 0.1).

Table 2: Level-2 discourse senses.

Value	Count
`Contrast`	17
`Evidence`	15
`Cause-Effect`	10
`Concession`	10
`Disjunction`	10
`Precedence`	10
`Restatement`	10
`Alternative`	9
`Effect-Cause`	5

Table 3: Level-1 couplings.

Value	Count
`entailment`	30
`contradiction`	27
`co_necessity`	10
`equivalence`	10
`none`	10
`exclusive`	9

Table 4: Relation validity.

Value	Count
`valid`	59
`invalid`	37

Table 5: Planted error kinds.

Value	Count
`spurious`	12
`false_endpoint`	10
`wrong_direction`	10
`wrong_sense`	5

37 of 96 gold relations (39%) are deliberately invalid: the corpus plants relation-level errors so a system's ability to *recover* the intended graph can be graded. The `gold` arm scores the graph as labelled, including those errors; the `gold_valid` arm scores only the intended-correct subgraph.

Table 6: LCS readouts, arm `gold` (all gold edges). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	**mean-marg**	**consist**	**reified**	**log-Z**	log Z
`f001`	0 (worse)	10	0.750	0.131	0.151	0.343	-8.01
`f001`	1 (base)	9	0.761	0.341	0.222	0.375	-6.93
`f001`	2 (concession_resolved)	9	0.790	0.085	0.253	0.374	-5.75
`f001`	3 (fix_one_conflict)	8	0.789	0.094	0.262	0.425	-5.07
`f001`	4 (coherent)	7	0.823	0.128	0.321	0.489	-3.59
`f002`	0 (worse)	10	0.786	0.016	0.120	0.281	-7.05
`f002`	1 (base)	9	0.792	0.075	0.250	0.327	-5.88
`f002`	2 (concession_resolved)	9	0.824	0.015	0.240	0.325	-4.84
`f002`	3 (fix_one_conflict)	8	0.824	0.016	0.248	0.393	-4.15
`f002`	4 (coherent)	7	0.824	0.018	0.257	0.500	-3.46

Table 7: LCS readouts, arm `gold_valid` (valid edges only). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	**mean-marg**	**consist**	**reified**	**log-Z**	log Z
`f001`	0 (worse)	5	0.749	0.450	0.334	0.432	-4.45
`f001`	1 (base)	5	0.749	0.450	0.334	0.432	-4.45
`f001`	2 (concession_resolved)	5	0.781	0.127	0.294	0.441	-3.51
`f001`	3 (fix_one_conflict)	5	0.781	0.127	0.294	0.441	-3.51
`f001`	4 (coherent)	4	0.814	0.173	0.357	0.514	-2.03
`f002`	0 (worse)	6	0.792	0.092	0.269	0.435	-3.76
`f002`	1 (base)	6	0.792	0.092	0.269	0.435	-3.76
`f002`	2 (concession_resolved)	6	0.824	0.018	0.258	0.498	-2.71
`f002`	3 (fix_one_conflict)	6	0.824	0.018	0.258	0.498	-2.71
`f002`	4 (coherent)	6	0.824	0.018	0.258	0.498	-2.71

Table 8: LCS readouts, arm `mined:llama-3.3-70b-instruct:windowed` (mined by llama-3.3-70b-instruct, windowed). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	**mean-marg**	**consist**	**reified**	**log-Z**	log Z
`f001`	0 (worse)	24	0.922	0.000	0.194	0.138	-5.41
`f001`	1 (base)	23	0.919	0.000	0.393	0.190	-4.33
`f001`	2 (concession_resolved)	22	0.905	0.000	0.328	0.135	-4.90
`f001`	3 (fix_one_conflict)	22	0.924	0.000	0.478	0.188	-4.05
`f001`	4 (coherent)	19	0.854	0.000	0.495	0.132	-6.77
`f002`	0 (worse)	17	0.855	0.000	0.492	0.067	-5.88
`f002`	1 (base)	19	0.678	0.054	0.314	0.166	-16.42
`f002`	2 (concession_resolved)	14	0.743	0.001	0.426	0.174	-8.05
`f002`	3 (fix_one_conflict)	14	0.744	0.016	0.406	0.172	-10.88
`f002`	4 (coherent)	15	0.785	0.000	0.230	0.232	-7.39

Table 9: LCS readouts, arm `mined:llama-3.3-70b-instruct:all_pairs` (mined by llama-3.3-70b-instruct, all pairs). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	**mean-marg**	**consist**	**reified**	**log-Z**	$\log Z$
`f001`	0 (worse)	97	0.789	0.012	0.000	0.007	-66.10
`f001`	1 (base)	87	0.262	0.341	0.018	0.007	-75.94
`f001`	2 (concession_resolved)	97	0.361	0.350	0.015	0.011	-83.03
`f001`	3 (fix_one_conflict)	95	0.570	0.023	0.197	0.041	-73.99
`f001`	4 (coherent)	96	0.735	0.068	0.097	0.076	-77.37
`f002`	0 (worse)	79	0.807	0.000	0.000	0.004	-71.97
`f002`	1 (base)	75	0.774	0.000	0.000	0.024	-70.41
`f002`	2 (concession_resolved)	85	0.672	0.256	0.001	0.004	-83.03
`f002`	3 (fix_one_conflict)	82	0.124	0.444	0.187	0.031	-83.11
`f002`	4 (coherent)	86	0.074	0.544	0.033	0.007	-79.19

3 LCS scores

3.1 Effect of the planted invalid relations

Removing the deliberately-invalid gold relations is the clearest single-number effect in this evaluation. The planted errors are predominantly conflict edges attached to a false endpoint, so deleting them removes active contradictions: the conflict-sensitive readouts move sharply, while `mean_marginal` — an average over atom marginals already pinned near their 0.9/0.1 priors — barely moves.

Table 10: Change in each readout from arm `gold` to arm `gold_valid` (valid edges only). Positive means the latter scores as more coherent.

Family	Rung	Rel	**mean-marg**	**consist**	**reified**	**log-Z**
`f001`	0	$10 \to 5$	-0.001	+0.318	+0.183	+0.090
`f001`	1	$9 \to 5$	-0.011	+0.109	+0.112	+0.058
`f001`	2	$9 \to 5$	-0.009	+0.042	+0.041	+0.067
`f001`	3	$8 \to 5$	-0.008	+0.033	+0.032	+0.016
`f001`	4	$7 \to 4$	-0.008	+0.046	+0.036	+0.026
`f002`	0	$10 \to 6$	+0.007	+0.076	+0.149	+0.153
`f002`	1	$9 \to 6$	-0.000	+0.017	+0.019	+0.107
`f002`	2	$9 \to 6$	-0.000	+0.003	+0.019	+0.173
`f002`	3	$8 \to 6$	+0.000	+0.002	+0.010	+0.105
`f002`	4	$7 \to 6$	+0.001	+0.000	+0.001	-0.002

3.2 Mined relations versus the gold labels

This is the mined-versus-labelled comparison: identical atoms, identical priors and identical readouts, with the relation graph the only thing that changed. A positive delta means the graph the miner recovered from the prose scores as *more* coherent than the one the corpus asserts — which, given the corpus deliberately plants conflict edges, is the expected direction whenever the miner misses them. Read it alongside the recall tables in Section 4: a large positive delta with low conflict recall means the readout moved because edges are missing, not because the text is more coherent.

Table 11: Change in each readout from arm `gold` to arm `mined:llama-3.3-70b-instruct:windowed` (mined by llama-3.3-70b-instruct, windowed). Positive means the latter scores as more coherent.

Family	Rung	Rel	**mean-marg**	**consist**	**reified**	**log-Z**
`f001`	0	10 → 24	+0.172	-0.131	+0.043	-0.205
`f001`	1	9 → 23	+0.159	-0.341	+0.171	-0.185
`f001`	2	9 → 22	+0.115	-0.085	+0.075	-0.239
`f001`	3	8 → 22	+0.135	-0.094	+0.217	-0.238
`f001`	4	7 → 19	+0.031	-0.128	+0.174	-0.356
`f002`	0	10 → 17	+0.069	-0.016	+0.372	-0.214
`f002`	1	9 → 19	-0.115	-0.021	+0.064	-0.161
`f002`	2	9 → 14	-0.081	-0.014	+0.187	-0.151
`f002`	3	8 → 14	-0.080	-0.000	+0.158	-0.221
`f002`	4	7 → 15	-0.038	-0.018	-0.027	-0.268

This is the mined-versus-labelled comparison: identical atoms, identical priors and identical readouts, with the relation graph the only thing that changed. A positive delta means the graph the miner recovered from the prose scores as *more* coherent than the one the corpus asserts — which, given the corpus deliberately plants conflict edges, is the expected direction whenever the miner misses them. Read it alongside the recall tables in Section 4: a large positive delta with low conflict recall means the readout moved because edges are missing, not because the text is more coherent.

Table 12: Change in each readout from arm `gold` to arm `mined:llama-3.3-70b-instruct:all_pairs` (mined by llama-3.3-70b-instruct, all pairs). Positive means the latter scores as more coherent.

Family	Rung	Rel	**mean-marg**	**consist**	**reified**	**log-Z**
`f001`	0	10 → 97	+0.039	-0.119	-0.150	-0.336
`f001`	1	9 → 87	-0.498	+0.000	-0.204	-0.367
`f001`	2	9 → 97	-0.428	+0.265	-0.238	-0.363
`f001`	3	8 → 95	-0.219	-0.071	-0.065	-0.384
`f001`	4	7 → 96	-0.088	-0.059	-0.224	-0.413
`f002`	0	10 → 79	+0.022	-0.016	-0.120	-0.278
`f002`	1	9 → 75	-0.019	-0.075	-0.250	-0.303
`f002`	2	9 → 85	-0.153	+0.241	-0.238	-0.321
`f002`	3	8 → 82	-0.700	+0.428	-0.061	-0.363
`f002`	4	7 → 86	-0.750	+0.526	-0.224	-0.493

4 Mining quality

A mined arm is scored on exactly the atoms and priors the gold arms use, so the tables below isolate relation mining. Two properties of the candidate-pair policies must be read alongside them, because both are definitional rather than empirical.

The policies do not have the same reach. `windowed` selects only *forward* pairs (source before target in atom order), refined by what the response actually links; `all_pairs` selects every one of the $n(n-1)$ ordered pairs. A gold relation that is *directed* and runs backward in atom order is therefore unreachable under `windowed` no matter what the prose says. Undirected couplings (equivalence, exclusive, co-necessity) are matched on the unordered pair, so they are not affected. Recall is reported split by direction below for exactly this reason: the backward-directed shortfall is a property of the policy, not a miner failure.

The policies do not build equally dense networks. `all_pairs` visits each unordered pair twice, once in each direction, and the graph builder does not deduplicate edges. A pair the model couples both ways therefore contributes *two* factors over the same two variables. The **dup** column counts this. Neither `windowed` (forward-only) nor gold (one relation per pair) can do it, so a comparison of LCS values across policies conflates coverage, directionality and network density — it is not a clean coverage ablation.

Table 13: Mined-versus-gold edge agreement, micro-averaged over items (percentages). **Pair** ignores the coupling and asks only whether the two atoms were related at all; **coupling** additionally requires the Level-1 coupling, with direction required only for the asymmetric ones, and is the level the MRF actually uses; **sense** additionally requires the Level-2 discourse sense. **Gold** counts only edge-producing relations. **dup** is unordered pairs related twice.

Policy	Mined	Gold	Pair P	Pair R	Coup P	Coup R	Sense R	dup
`windowed`	189	86	36.5	80.2	24.9	54.7	43.0	0
`all_pairs`	879	86	14.8	98.8	9.1	81.4	58.1	306

Table 14: Coupling-level recall (%) stratified by the gold edge's own properties. **Fwd/Bwd** is the sign of the target-minus-source atom index; **Dir/Undir** is whether the coupling is asymmetric; **win/gate** is the generator's own `window_admission` label, i.e. whether it expected the edge inside a radius-4 window. A forward-only policy is structurally capped in the **Bwd**×**Dir** cell.

Policy	Fwd	Bwd	Dir	Undir	win	gate
`windowed`	73.2	20.0	25.0	80.4	58.8	0.0
`all_pairs`	83.9	76.7	80.0	82.6	81.2	83.3

Table 15: Coupling-level recall (%) on valid versus deliberately-invalid gold edges, and violations of the declared non-relations. The **invalid** column reads *inverted*: those edges are planted errors and the miner reads the response, so failing to recover them is arguably correct — lower is better. **Non-rel** counts pairs the item explicitly declares unrelated that the miner nevertheless coupled; unlike precision against all unlabelled pairs, this denominator is a real negative set.

Policy	valid R	invalid R	Non-rel	rate
`windowed`	66.7	34.4	7/48	14.6
`all_pairs`	87.0	71.9	13/48	27.1

One confound to note in Table 14: in this dataset every `gate`-admitted gold edge is also marked `invalid`, so the **gate** column and the **invalid** column of Table 15 are not independent.

5 Arm comparison

Table 16: Each readout averaged over items, per arm. Averaging across the rungs of a ladder deliberately collapses the ordering the ladder is built to test, so this table is a level comparison only; the ordering question is Table 17.

Arm	Cells	**mean-marg**	**consist**	**reified**	**log-Z**
`gold`	10	0.796	0.092	0.232	0.383
`gold_valid`	10	0.793	0.157	0.292	0.462
`mined:llama-3.3-70b-instruct:windowed`	10	0.833	0.007	0.376	0.159
`mined:llama-3.3-70b-instruct:all_pairs`	10	0.517	0.204	0.055	0.021

Table 17: Ordering constraints satisfied per arm. With 28 assertions per arm this is the only comparison here with enough observations to be more than anecdote: it asks whether the readouts order the rungs correctly, which is what the ladder was built to test.

Arm	`f001`	`f002`	total	%
`gold`	11/14	11/14	22/28	78.6
`gold_valid`	7/14	4/14	11/28	39.3
`mined:llama-3.3-70b-instruct:windowed`	3/14	9/14	12/28	42.9
`mined:llama-3.3-70b-instruct:all_pairs`	10/14	5/14	15/28	53.6

6 Ladder ordering constraints

Each family declares the ordering its rungs are meant to exhibit; the graded readouts are `mean_marginal`, `consistency`, `log_partition`.

Every rung carries its own gold relations. In each family the five rungs carry *different* gold relation sets, so each rung's labels describe the response that ships with them and the ordering constraints below are a genuine test of the readouts rather than a measurement of the corpus. This is what the per-rung relation transforms in `locobench.perturb` (`apply_calls` / `plan_targets`) provide: a rung's edges are the base edge set with that rung's own perturbations applied, each call targeting a distinct eligible edge, and a rung whose edge set still matches its parent's while its calls claim a change is rejected at generation time.

Read the two arms differently. A `fix_one_conflict` or `coherent` rung removes the *planted-invalid* conflicts first — those are the edges a fix most plausibly deletes. The `gold_valid` arm has already discarded exactly those edges, so consecutive rungs can share an identical valid subgraph and that arm is close to flat by construction. Its strict-increase failures are therefore expected and carry no information about the readouts; **the `gold` arm is the one the ladder constraints are about**. The valid-only arm remains useful for what it was added for: quantifying what the planted errors cost at a fixed rung.

Family `f001` (Anthropology, CONFLICT)

Gold relations identical across rungs: **no**; distinct response texts: 5 of 5.

Table 18: Ordering constraints for `f001`, arm `gold`: 11 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	increase	0.010	✓
C1	consist	0→1	increase	increase	0.209	✓
C1	log-Z	0→1	increase	increase	0.032	✓
C1	mean-marg	1→2	increase	increase	0.029	✓
C1	mean-marg	2→3	increase	decrease	-0.000	×
C1	log-Z	2→3	increase	increase	0.051	✓
C1	mean-marg	3→4	increase	increase	0.034	✓
C1	consist	3→4	increase	increase	0.034	✓
C1	log-Z	3→4	increase	increase	0.063	✓
C2	consist	1→2	decrease	decrease	-0.256	✓
C2	log-Z	1→2	invariant	decrease	-0.000	×
C3	mean-marg	0→4	increase	increase	0.072	✓
C3	consist	0→4	increase	decrease	-0.004	×
C3	log-Z	0→4	increase	increase	0.146	✓

Table 19: Ordering constraints for `f001`, arm `gold_valid`: 7 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	invariant	0.000	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	invariant	0.000	×
C1	mean-marg	1→2	increase	increase	0.032	✓
C1	mean-marg	2→3	increase	invariant	0.000	×
C1	log-Z	2→3	increase	invariant	0.000	×
C1	mean-marg	3→4	increase	increase	0.034	✓
C1	consist	3→4	increase	increase	0.046	✓
C1	log-Z	3→4	increase	increase	0.073	✓
C2	consist	1→2	decrease	decrease	-0.323	✓
C2	log-Z	1→2	invariant	increase	0.009	×
C3	mean-marg	0→4	increase	increase	0.065	✓
C3	consist	0→4	increase	decrease	-0.277	×
C3	log-Z	0→4	increase	increase	0.082	✓

Table 20: Ordering constraints for `f001`, arm `mined:llama-3.3-70b-instruct:windowed`: 3 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.003	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	increase	0.052	✓
C1	mean-marg	1→2	increase	decrease	-0.014	×
C1	mean-marg	2→3	increase	increase	0.019	✓
C1	log-Z	2→3	increase	increase	0.053	✓
C1	mean-marg	3→4	increase	decrease	-0.070	×
C1	consist	3→4	increase	invariant	0.000	×
C1	log-Z	3→4	increase	decrease	-0.055	×
C2	consist	1→2	decrease	invariant	0.000	×
C2	log-Z	1→2	invariant	decrease	-0.055	×
C3	mean-marg	0→4	increase	decrease	-0.068	×
C3	consist	0→4	increase	invariant	0.000	×
C3	log-Z	0→4	increase	decrease	-0.005	×

Table 21: Ordering constraints for `f001`, arm `mined:llama-3.3-70b-instruct:all_pairs`: 10 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.527	×
C1	consist	0→1	increase	increase	0.329	✓
C1	log-Z	0→1	increase	increase	0.001	✓
C1	mean-marg	1→2	increase	increase	0.099	✓
C1	mean-marg	2→3	increase	increase	0.209	✓
C1	log-Z	2→3	increase	increase	0.030	✓
C1	mean-marg	3→4	increase	increase	0.165	✓
C1	consist	3→4	increase	increase	0.045	✓
C1	log-Z	3→4	increase	increase	0.035	✓
C2	consist	1→2	decrease	increase	0.009	×
C2	log-Z	1→2	invariant	increase	0.004	×
C3	mean-marg	0→4	increase	decrease	-0.054	×
C3	consist	0→4	increase	increase	0.056	✓
C3	log-Z	0→4	increase	increase	0.069	✓

Family `f002` (Archaeology, CONFLICT)

Gold relations identical across rungs: **no**; distinct response texts: 5 of 5.

Table 22: Ordering constraints for `f002`, arm `gold`: 11 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	`mean-marg`	0→1	increase	increase	0.007	✓
C1	`consist`	0→1	increase	increase	0.059	✓
C1	`log-Z`	0→1	increase	increase	0.046	✓
C1	`mean-marg`	1→2	increase	increase	0.032	✓
C1	`mean-marg`	2→3	increase	decrease	-0.000	×
C1	`log-Z`	2→3	increase	increase	0.069	✓
C1	`mean-marg`	3→4	increase	decrease	-0.000	×
C1	`consist`	3→4	increase	increase	0.002	✓
C1	`log-Z`	3→4	increase	increase	0.106	✓
C2	`consist`	1→2	decrease	decrease	-0.061	✓
C2	`log-Z`	1→2	invariant	decrease	-0.002	×
C3	`mean-marg`	0→4	increase	increase	0.038	✓
C3	`consist`	0→4	increase	increase	0.002	✓
C3	`log-Z`	0→4	increase	increase	0.218	✓

Table 23: Ordering constraints for `f002`, arm `gold_valid`: 4 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	`mean-marg`	0→1	increase	invariant	0.000	×
C1	`consist`	0→1	increase	invariant	0.000	×
C1	`log-Z`	0→1	increase	invariant	0.000	×
C1	`mean-marg`	1→2	increase	increase	0.032	✓
C1	`mean-marg`	2→3	increase	invariant	0.000	×
C1	`log-Z`	2→3	increase	invariant	0.000	×
C1	`mean-marg`	3→4	increase	invariant	0.000	×
C1	`consist`	3→4	increase	invariant	0.000	×
C1	`log-Z`	3→4	increase	invariant	0.000	×
C2	`consist`	1→2	decrease	decrease	-0.074	✓
C2	`log-Z`	1→2	invariant	increase	0.063	×
C3	`mean-marg`	0→4	increase	increase	0.032	✓
C3	`consist`	0→4	increase	decrease	-0.074	×
C3	`log-Z`	0→4	increase	increase	0.063	✓

7 Worked examples

Each example gives the response as generated, its atoms with the priors their `factual` labels induce, the relation graph the gold labels build, and the specific relations with the factor probability each contributed.

Table 24: Ordering constraints for `f002`, arm `mined:llama-3.3-70b-instruct:windowed`: 9 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.177	×
C1	consist	0→1	increase	increase	0.054	✓
C1	log-Z	0→1	increase	increase	0.099	✓
C1	mean-marg	1→2	increase	increase	0.066	✓
C1	mean-marg	2→3	increase	increase	0.000	✓
C1	log-Z	2→3	increase	decrease	-0.002	×
C1	mean-marg	3→4	increase	increase	0.042	✓
C1	consist	3→4	increase	decrease	-0.016	×
C1	log-Z	3→4	increase	increase	0.060	✓
C2	consist	1→2	decrease	decrease	-0.053	✓
C2	log-Z	1→2	invariant	increase	0.008	×
C3	mean-marg	0→4	increase	decrease	-0.069	×
C3	consist	0→4	increase	increase	0.000	✓
C3	log-Z	0→4	increase	increase	0.165	✓

Table 25: Ordering constraints for `f002`, arm `mined:llama-3.3-70b-instruct:all_pairs`: 5 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.033	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	increase	0.020	✓
C1	mean-marg	1→2	increase	decrease	-0.102	×
C1	mean-marg	2→3	increase	decrease	-0.548	×
C1	log-Z	2→3	increase	increase	0.027	✓
C1	mean-marg	3→4	increase	decrease	-0.050	×
C1	consist	3→4	increase	increase	0.100	✓
C1	log-Z	3→4	increase	decrease	-0.024	×
C2	consist	1→2	decrease	increase	0.256	×
C2	log-Z	1→2	invariant	decrease	-0.020	×
C3	mean-marg	0→4	increase	decrease	-0.733	×
C3	consist	0→4	increase	increase	0.544	✓
C3	log-Z	0→4	increase	increase	0.003	✓

7.1 Anthropology – base (`locobench-claude-5-fixed-f001-r1`)

Topic: Anthropology (Social Sciences). **Family:** `f001` (CONFLICT), rung 1 (*base*). **Scores:** mean-marg = 0.761, consist = 0.341, reified = 0.222, log-Z = 0.375.

Framing question. *How do anthropologists adjudicate between rival interpretations when skeletal remains at a burial site show signs of trauma, and what evidence establishes whether the injuries were inflicted by human violence or by post-mortem processes?*

Response. Adjudication between rival readings of a traumatized skeleton begins as a matter of procedure rather than interpretation, because the evidentiary value of a lesion depends on how much is known about its context. Excavators record the position and orientation of each bone in situ before the remains are lifted, and only once that mapping is complete do several independent osteologists score the same lesions to test inter-observer reliability. Ordering the work in this way ensures that the descriptive record is fixed while the burial is still articulated, and that the scoring of margins, depressions and fracture surfaces can later be checked against a documented spatial arrangement rather than against the recollection of whoever first noticed the damage.

The analytical leverage in such disputes comes from the mechanics of bone itself, which change as the tissue's organic fraction degrades. The loss of collagen after death shifts the bone's response to loading from plastic to brittle, so that the same blow yields quite different failure geometry depending on when it was delivered. By contrast, buried bone retains its fresh, plastic mechanical properties for approximately fifty years after death, a durability that determines how wide the diagnostic window for early damage can reasonably be drawn.

Those material states leave characteristic signatures on the break surface, and it is these that carry most of the interpretive weight. Perimortem fractures in fresh bone produce beveled or obliquely angled fracture margins, whereas postmortem breaks in dry bone typically show transverse edges meeting the cortical surface at right angles; indeed, the plastic-to-brittle shift in loading response noted above is itself a consequence of that right-angled transverse geometry. The picture is complicated by the fact that postmortem dry-bone fractures produce beveled margins with adhering flakes of cortical bone. Even so, adhering bone flakes and hinge fractures indicate the bone was fresh when it broke, and analysts continue to treat that combination as the most reliable single criterion available to them.

Applied to a contested specimen, the reasoning runs from timing to morphology. The fracture occurred while the bone was still fresh and elastic, which confirms that flaking and hinging are markers of fresh-bone breakage; in other words, the break happened at or near the time of death, before the bone's collagen degraded. At the level of the whole assemblage, the dispute reduces to two mutually exclusive readings, exactly one of which must be right: either every fracture margin on the skeleton was produced after death, or at least one fracture margin on the skeleton was produced at or around the time of death. Faced with precisely this choice in the case at hand, the review panel ultimately held that beveled fracture margins on the cranium reflect perimortem breakage, a determination that dissolved the apparent conflict between the two accounts of beveling by fixing which of them governs the cranial evidence.

Establishing that a break is early does not by itself establish human violence, since several agencies operate on a body in the ground and each leaves its own configuration of displacement and damage. Ranking those agencies against the field documentation is the final step. For this assemblage, at least one of two processes was at work, and perhaps both: scavenging animals disturbed the remains within the grave, or sediment loading fractured the bones after burial. Distinguishing them relies on the recorded orientation of elements, the distribution of tooth pitting and the compression geometry of the affected shafts. A further source of confusion arises from the recovery process itself, for damage inflicted by trowels and picks during excavation can mimic ancient trauma; such tool damage is what caused the post-burial fracturing under sediment load in these bones, and its recognition is therefore a precondition of any claim about interpersonal violence.

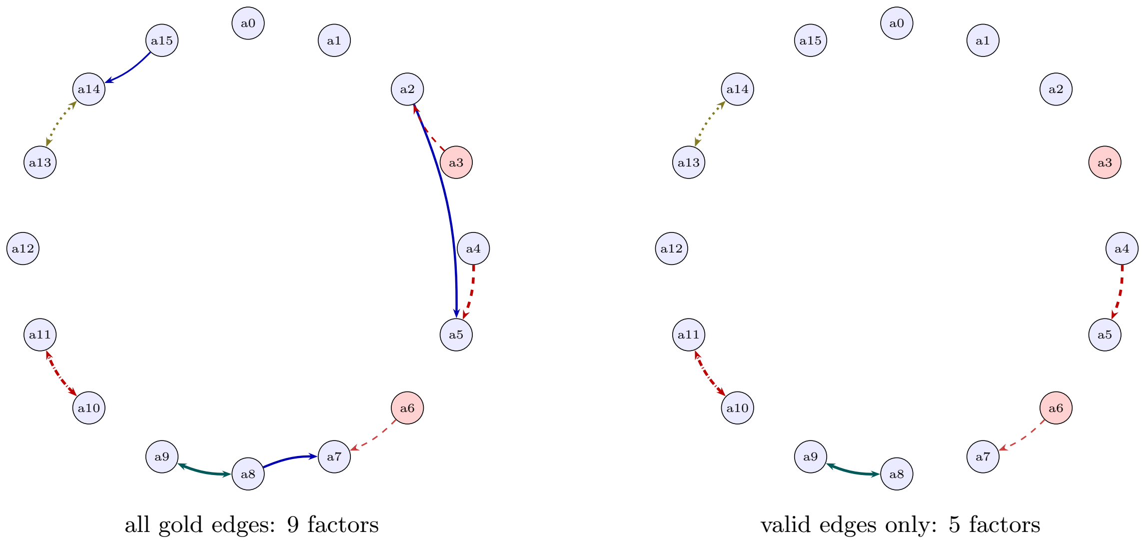


Figure 1: Relation graph for `locobench-claude-5-fixed-f001-r1`. Nodes are atoms on a circle, tinted red when the item marks them non-factual (prior 0.1) and blue when factual (prior 0.9). Edges are gold relations: solid blue = entailment, dashed red = contradiction, double-headed teal = equivalence, dash-dotted red (double-headed) = exclusive (exactly-one), dotted olive (double-headed) = co-necessity (at-least-one); thickness scales with the factor probability. Ordering-only Precedence edges carry no factor and so are not drawn.

Table 26: The specific gold relations of `locobench-claude-5-fixed-f001-r1`. p **used** is the factor probability the MRF received: the band midpoint, times $(1-\lambda)$ for a resolved concession, or *none* for an ordering-only edge (no factor). **Valid** marks the deliberately-planted errors with their kind. **Res.** is the resolving atom of a resolved concession.

ID	Endpoints	Sense	Coupling	Band	p used	Valid	Res.
`r000`	`a0` → `a1`	Precedence	`none`	moderate	*none*	✓	
`r001`	`a2` → `a5`	Effect-Cause	`entailment`	moderate	0.720	× wrong_direction	
`r002`	`a3` ↔ `a2`	Contrast	`contradiction`	weak	0.470	× false_endpoint	
`r003`	`a4` ↔ `a5`	Contrast	`contradiction`	strong	0.925	✓	
`r004`	`a6` → `a7`	Concession	`contradiction`	moderate	0.396	✓	`a12`
`r005`	`a8` → `a7`	Evidence	`entailment`	moderate	0.720	× wrong_direction	
`r006`	`a8` ↔ `a9`	Restatement	`equivalence`	strong	0.925	✓	
`r007`	`a10` ↔ `a11`	Alternative	`exclusive`	strong	0.925	✓	
`r008`	`a13` ↔ `a14`	Disjunction	`co_necessity`	moderate	0.720	✓	
`r009`	`a15` → `a14`	Cause-Effect	`entailment`	weak	0.470	× spurious	

Table 27: Atoms of `locobench-claude-5-fixed-f001-r1`: the `factual` label, the prior it maps to, the posterior marginal $P(a_i{=}1)$ the MRF returns, and the shift. Atoms dragged below their prior are the ones the coherence structure penalises.

Atom	fact	π_i	$P(a_i{=}1)$	Δ	Text
`a0`	✓	0.90	0.900	+0.000	Excavators record the position and orientation of each bone in situ before the remains are lifted.
`a1`	✓	0.90	0.900	+0.000	Several independent osteologists score the same lesions to test inter-observer reliability.
`a2`	✓	0.90	0.904	+0.004	The loss of collagen after death shifts the bone's response to loading from plastic to brittle.
`a3`	×	0.10	0.104	+0.004	Buried bone retains its fresh, plastic mechanical properties for approximately fifty years after death.
`a4`	✓	0.90	0.672	-0.228	Perimortem fractures in fresh bone produce beveled or obliquely angled fracture margins.
`a5`	✓	0.90	0.735	-0.165	Postmortem breaks in dry bone typically show transverse edges meeting the cortical surface at right angles.
`a6`	×	0.10	0.117	+0.017	Postmortem dry-bone fractures produce beveled margins with adhering flakes of cortical bone.
`a7`	✓	0.90	0.959	+0.059	Adhering bone flakes and hinge fractures indicate the bone was fresh when it broke.
`a8`	✓	0.90	0.985	+0.085	The fracture occurred while the bone was still fresh and elastic.
`a9`	✓	0.90	0.982	+0.082	The break happened at or near the time of death, before the bone's collagen degraded.
`a10`	✓	0.90	0.632	-0.268	Every fracture margin on the skeleton was produced after death.
`a11`	✓	0.90	0.632	-0.268	At least one fracture margin on the skeleton was produced at or around the time of death.
`a12`	✓	0.90	0.900	+0.000	The review panel ultimately held that beveled fracture margins on the cranium reflect perimortem breakage.
`a13`	✓	0.90	0.929	+0.029	Scavenging animals disturbed the remains within the grave.
`a14`	✓	0.90	0.922	+0.022	Sediment loading fractured the bones after burial.
`a15`	✓	0.90	0.895	-0.005	Damage inflicted by trowels and picks during excavation can mimic ancient trauma.

7.2 Archaeology – base (`locobench-claude-5-fixed-f002-r1`)

Topic: Archaeology (Humanities). **Family:** `f002` (CONFLICT), rung 1 (*base*). **Scores:** mean-marg = 0.792, consist = 0.075, reified = 0.250, log-Z = 0.327.

Framing question. *How do archaeologists adjudicate between competing interpretations when a stratified deposit yields artifacts that appear older than the layer containing them?*

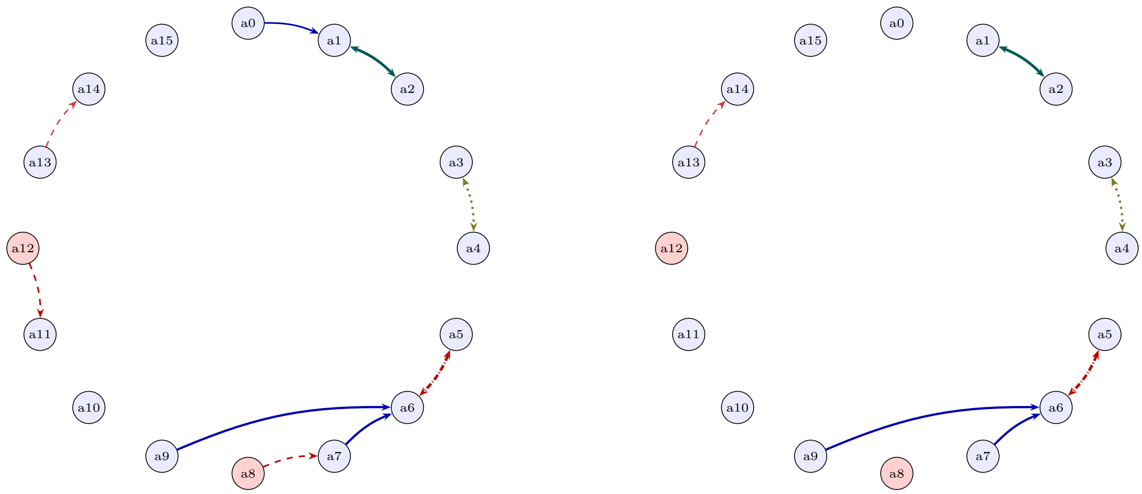
Response. When a stratified deposit yields material that appears to predate the layer holding it, the first task is not to discard either observation but to specify what each rests on. The law of superposition holds that in an undisturbed sequence lower deposits were laid down before those above them, and it was on the strength of that principle that the artifacts proved older than the sediment enclosing them. Chronological information at a site is carried in two partly independent registers — the order of the deposits and the age of the objects within them — and when those registers disagree the disagreement is informative rather than fatal. The same discrepancy can be stated from the other end without any change in content: equivalently, the sediment enclosing the artifacts was younger than the artifacts. What remains to be settled is which of the several histories capable of producing such an arrangement actually did so.

Two families of process can place old objects in young sediment, and they are not mutually exclusive, so at least one of the following obtained, perhaps both: older material was incorporated into the layer while the layer was forming, or older artifacts moved into the layer after the layer had formed. Discriminating between them requires a prior judgment about the integrity of the deposit, and there the options are exhaustive and mutually exclusive: either the deposit containing the artifacts was stratigraphically intact, or it was disturbed after its formation, and since both cannot be true, one of them must hold. That burrowing animals can transport artifacts vertically between stratigraphic layers indicates the second of those alternatives, post-depositional disturbance, wherever traces of such activity can be recognized in section. By contrast with that burrowing transport, bioturbation displaces artifacts downward only.

General mechanisms, however, license possibilities rather than conclusions; adjudication is achieved by site-specific argument, marshalled from the excavated record itself. Here the final excavation report concluded that the deposit had been disturbed by rodent burrowing after its formation, and that determination constitutes the strongest evidence that the deposit's stratigraphic integrity had in fact been compromised. Refitting fragments of a single broken artifact across two layers demonstrates vertical displacement of material, and that demonstration was accomplished earlier than the conclusion eventually recorded in the report, so the argument for movement between strata was already in hand when the disturbance was formally diagnosed.

Independent chronometry supplies a second line of control on the same question. Radiocarbon dating of short-lived plant remains from a context provides an age estimate for that context's formation, because seeds, twigs and other ephemeral tissues enter a deposit close to the moment it accumulates and cannot have been curated for generations beforehand. Radiocarbon dating directly measures the age at which a flint tool was knapped; the plant-based estimate, by contrast, speaks to the accumulation of the enclosing matrix rather than to the objects recovered from within it, and the two therefore answer different questions about the same anomaly.

Where rival readings survive these stratigraphic and chronometric controls, resolution becomes a matter of specialist review. Although the excavator interpreted the older flint blades as heirlooms curated by the occupants of the layer, a lithics specialist interpreted those same blades as residual material eroded from an earlier deposit, reading their condition and technological character as products of reworking rather than of deliberate retention. The impasse was closed when a review panel determined that the older flint blades were residual finds rather than curated heirlooms, a finding that removed the curation hypothesis from consideration and aligned the artifact assemblage with the depositional history reconstructed from the section. Adjudication, in short, proceeds by converting an apparent contradiction into a testable choice among formation processes, and then by letting stratigraphic observation, refitting, dating and peer scrutiny narrow that choice to one.



all gold edges: 9 factors

valid edges only: 6 factors

Figure 2: Relation graph for `locobench-claude-5-fixed-f002-r1`. Nodes are atoms on a circle, tinted red when the item marks them non-factual (prior 0.1) and blue when factual (prior 0.9). Edges are gold relations: solid blue = entailment, dashed red = contradiction, double-headed teal = equivalence, dash-dotted red (double-headed) = exclusive (exactly-one), dotted olive (double-headed) = co-necessity (at-least-one); thickness scales with the factor probability. Ordering-only Precedence edges carry no factor and so are not drawn.

Table 28: The specific gold relations of `locobench-claude-5-fixed-f002-r1`. p **used** is the factor probability the MRF received: the band midpoint, times $(1-\lambda)$ for a resolved concession, or *none* for an ordering-only edge (no factor). **Valid** marks the deliberately-planted errors with their kind. **Res.** is the resolving atom of a resolved concession.

ID	Endpoints	Sense	Coupling	Band	p used	Valid	Res.
`r000`	`a0` $\rightarrow$ `a1`	Cause-Effect	`entailment`	weak	0.470	× spurious	
`r001`	`a1` $\leftrightarrow$ `a2`	Restatement	`equivalence`	strong	0.925	✓	
`r002`	`a3` $\leftrightarrow$ `a4`	Disjunction	`co_necessity`	moderate	0.720	✓	
`r003`	`a5` $\leftrightarrow$ `a6`	Alternative	`exclusive`	strong	0.925	✓	
`r004`	`a7` $\rightarrow$ `a6`	Evidence	`entailment`	moderate	0.720	✓	
`r005`	`a8` $\leftrightarrow$ `a7`	Contrast	`contradiction`	weak	0.470	× false_endpoint	
`r006`	`a9` $\rightarrow$ `a6`	Evidence	`entailment`	moderate	0.720	✓	
`r007`	`a10` $\rightarrow$ `a9`	Precedence	`none`	weak	*none*	× wrong_sense	
`r008`	`a12` $\leftrightarrow$ `a11`	Contrast	`contradiction`	weak	0.470	× false_endpoint	
`r009`	`a13` $\rightarrow$ `a14`	Concession	`contradiction`	moderate	0.396	✓	`a15`

Table 29: Atoms of `locobench-claude-5-fixed-f002-r1`: the `factual` label, the prior it maps to, the posterior marginal $P(a_i{=}1)$ the MRF returns, and the shift. Atoms dragged below their prior are the ones the coherence structure penalises.

Atom	fact	π_i	$P(a_i{=}1)$	Δ	Text
a0	✓	0.90	0.895	-0.005	The law of superposition holds that in an undisturbed sequence lower deposits were laid down before those above them.
a1	✓	0.90	0.977	+0.077	The artifacts were older than the sediment that enclosed them.
a2	✓	0.90	0.978	+0.078	The sediment enclosing the artifacts was younger than the artifacts.
a3	✓	0.90	0.929	+0.029	Older material was incorporated into the layer while the layer was forming.
a4	✓	0.90	0.929	+0.029	Older artifacts moved into the layer after the layer had formed.
a5	✓	0.90	0.478	-0.422	The deposit containing the artifacts was stratigraphically intact.
a6	✓	0.90	0.902	+0.002	The deposit containing the artifacts was disturbed after its formation.
a7	✓	0.90	0.920	+0.020	Burrowing animals can transport artifacts vertically between stratigraphic layers.
a8	×	0.10	0.105	+0.005	Bioturbation displaces artifacts downward only.
a9	✓	0.90	0.919	+0.019	The final excavation report concluded that the deposit had been disturbed by rodent burrowing after its formation.
a10	✓	0.90	0.900	+0.000	Refitting fragments of a single broken artifact across two layers demonstrates vertical displacement of material.
a11	✓	0.90	0.901	+0.001	Radiocarbon dating of short-lived plant remains from a context provides an age estimate for that context's formation.
a12	×	0.10	0.104	+0.004	Radiocarbon dating directly measures the age at which a flint tool was knapped.
a13	✓	0.90	0.913	+0.013	The excavator interpreted the older flint blades as heirlooms curated by the occupants of the layer.
a14	✓	0.90	0.929	+0.029	A lithics specialist interpreted the older flint blades as residual material eroded from an earlier deposit.
a15	✓	0.90	0.900	+0.000	A review panel determined that the older flint blades were residual finds rather than curated heirlooms.

8 Baseline reproduction

All **80** readout values shared with `results/locobench_claude_5_fixed_lcs` reproduce to within $0.0e+00$ (tolerance $1e-06$). The arms added by this revision therefore leave the previously-published gold numbers unchanged.

9 Findings

- **Every rung carries its own gold relations, so the ladder is a real test.** On the `gold` arm 22 of 28 ordering assertions hold across all families (`consist` 7/8, `log-Z` 8/10, `mean-marg` 7/10). Because each rung's labels describe the response that ships with them, these pass rates are evidence about the readouts rather than about the corpus — which is what the previous generation of this dataset could not provide.

- **The readouts differ sharply in sensitivity.** Across the 10 scored items `mean_marginal` spans only $[0.750, 0.824]$, while `consistency` spans $[0.015, 0.341]$. With atom priors pinned at 0.9/0.1 the mean marginal is dominated by those priors and moves little; the conflict-sensitive readouts are what register the contradiction structure. For a benchmark whose families vary conflict structure, `consistency` and `log_partition` are the discriminating readouts.

- **The planted invalid relations carry most of the incoherence.** Dropping them changes `consistency` by 0.000 to 0.318 across items. The planted errors are largely conflict edges anchored to a false endpoint, so they add active contradictions, and the intended-correct subgraph is markedly more coherent. That the effect runs in this direction is a sanity check that the gold labels and the MRF encoding agree about which edges are the damaging ones.

- **Relation counts and factor counts are not the same number.** Of 96 gold relations, 10 are Precedence edges that couple no truth values and never reach the MRF, and 37 are deliberately invalid. Any comparison of graph density needs both figures; Table 1 reports them separately for that reason.

- **Recovering the graph is harder than scoring it.** The best mined arm (`all_pairs`) reaches 98.8% recall at the pair level but only 81.4% once the Level-1 coupling has to match, at 9.1% precision. The gold arms show the readouts behave correctly on a correct graph; these numbers are what the pipeline actually delivers from prose, and the gap between the two is the finding.

- **The direction split is a policy property, not a model property.** `windowed` 73.2% forward vs 20.0% backward; `all_pairs` 83.9% forward vs 76.7% backward. A forward-only policy cannot emit a backward ordered pair at all, so its backward recall is bounded by the undirected couplings it can still match on the unordered pair. Any comparison of the two policies' coupling recall is therefore partly a comparison of their reach.

- **One policy scores a denser network than the others.** Duplicate unordered pairs: `windowed` 0, `all_pairs` 306. Because edges are not deduplicated, each duplicate puts a second factor on the same variable pair, so that pair's influence is applied twice. Readout differences between the policies cannot be attributed to coverage alone.

- **The declared negatives are a cleaner precision signal than the unlabelled pairs.** Violations of the item's own `non_relations`: `windowed` 7/48, `all_pairs` 13/48. Precision against every unlabelled pair is dominated by pairs of unknown status, whereas these are pairs the corpus asserts are unrelated.

- **The type confidence is saturated, so the factor probability is carrying only the strength.** 1017 of 1068 mined relations (95.2%) have $P(\tau \mid a_i, a_j) = 1.0$, with a minimum of 0.679 across the whole sweep. Because $p = P(\tau \mid a_i, a_j) \times P(a_j \mid a_i, \tau)$, the edge weight is effectively the conditional strength alone. This is a genuine reading and not a missing-logprobs fallback — that path returns 0.5, not 1.0 — but it means Prompt A contributes a label and no usable uncertainty on this model, and the two-factor decomposition is not being exercised. A model whose `[coupling=...]` span is less peaked, or a calibrator fitted on that span, would be needed to test it.

- **The mined numbers are not built on dropped calls.** 0 of 3732 LLM calls failed across the mined arms. This matters because a failed call is parsed as *no relation* — indistinguishable from a genuine negative — so an unmeasured failure rate would silently depress recall and inflate every readout. The run refuses a cell whose failure rate exceeds the configured ceiling.

10 Threats to validity

- **Gold relations are labels, not measurements.** The factor probability is a band midpoint, so a `strong` edge enters at exactly 0.925 whatever the underlying text supports. The gold arms therefore evaluate the MRF encoding and the readouts *given* perfect relation labels. They are the reference point the mined arms are measured against, not a mining result in themselves.

- **Two families is a small sample.** The dataset has 2 families over 10 items, all generated by one model. The spreads quoted above describe this dataset; no significance claim is made or implied.

- **The ladder tests the readouts, not the miner.** Gold relations vary per rung, so the ordering constraints are a real test — but of the MRF encoding and the readouts given perfect labels. A live system also has to *recover* those relations from the text, and nothing here measures that.

- **The concession discount is applied from the label.** Using the item's `resolver_atom_id` bypasses the text heuristic a live miner must use, so the resolved-concession behaviour shown here is the best case for that mechanism.

- **Inference is approximate in general.** Merlin runs weighted mini-bucket at a finite i-bound. On these 16-atom networks the induced width is small enough that it reports exact inference, but the normalized log-partition also depends on a MAP floor and a contradiction-free ceiling, so it is the readout most sensitive to any approximation.

- **The policy comparison is not a coverage ablation.** `windowed` and `all_pairs` differ in three ways at once: which pairs they reach, whether they consider both directions, and — because duplicate edges are not merged — how dense a network they build. A readout difference between them cannot be attributed to coverage alone. Isolating coverage would need one policy at two window radii.

- **A failed call and a genuine negative look the same.** The miner parses a captured exception as *no relation*. This run counts those failures and refuses a cell that exceeds a threshold, so the reported numbers are not built on dropped calls — but the underlying ambiguity is a property of the pipeline, and a run without that accounting would show no symptom.
- **One mined observation per cell.** Mining is sampled once per (item, arm) at the model's default temperature, so no run-to-run variance is measured. Differences between mined arms of a similar size should not be read as reliable.